

Wireless Power Transfer via Inductive Coupling: Harnessing Magnetic Fields for Contactless Energy Transmission

ELECENG 2FH4 - Bonus Research Assignment

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1. Introduction

Wireless power transfer (WPT) has become a mainstream technology, powering smartphones, medical implants, and electric vehicles. At its core lies inductive coupling, a direct manifestation of Faraday's Law and Ampère's Law. This write-up explains how time-varying magnetic fields enable contactless energy transfer, the mathematical framework governing the process, and practical engineering trade-offs.

2. Physical Concept: Mutual Induction

A time-varying current in a primary coil produces a time-varying magnetic flux. When a secondary coil is placed in this alternating flux, an electromotive force (EMF) is induced in the secondary coil. If the secondary circuit is closed, current flows and power is delivered to a load. Energy transfers via the magnetic field as an intermediary, requiring no physical connection. The efficiency depends on how well the magnetic field lines from the primary link with the secondary, quantified by mutual inductance M .

3. Mathematical Formulation

For two coils with turns N_1 and N_2 , mutual inductance M (in Henries) is:

$$M = \frac{N_2 \Phi_{21}}{I_1} = \frac{N_1 \Phi_{12}}{I_2}$$

where Φ_{21} is the flux through coil 2 due to current I_1 in coil 1. For coaxial circular coils of radii r_1, r_2 separated by distance d (with $d \gg r$):

$$M \approx \frac{\mu_0 \pi N_1 N_2 r_1^2 r_2^2}{2d^3}$$

The rapid $1/d^3$ drop-off explains why inductive WPT is efficient only over short distances (cm to a few meters).

The fraction of flux linked is the coupling coefficient k ($0 \leq k \leq 1$):

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

where L_1, L_2 are self-inductances. Perfect coupling ($k = 1$) means all primary flux passes through the secondary. In practice, k is 0.1–0.8 for loosely coupled systems (e.g., smartphone pads) and >0.9 for tightly coupled transformers.

Faraday's Law gives the open-circuit voltage in the secondary:

$$V_{oc} = -M \frac{dI_1}{dt}$$

For sinusoidal operation at angular frequency $\omega = 2\pi f$ with primary RMS current $I_{1,\text{rms}}$:

$$V_{2,\text{rms}} = \omega M I_{1,\text{rms}}$$

Key insight: Higher frequency increases induced voltage for a given M . Practical wireless chargers operate from 100 kHz (Qi standard) to several MHz (resonant systems).

The reflected impedance from secondary to primary is:

$$Z_{\text{ref}} = \frac{\omega^2 M^2}{R_L + j\omega L_2 + \frac{1}{j\omega C_2}}$$

At resonance, capacitive reactance cancels inductive reactance, making the impedance purely resistive and maximizing power transfer. Adding capacitors to both coils (series or parallel) creates a resonant transformer, boosting efficiency even when k is small.

4. Practical Relevance and Trade-offs

Applications: Smartphones (Qi: 15W, 80-90% at 5 mm); medical implants (low heating); EVs (3-20 kW, 90-95% at 150-200 mm); industrial (rotating machinery).

Key challenges:

- Distance: $M \propto 1/d^3$ causes rapid efficiency drop. Resonance and high- Q coils help.
- Misalignment: Reduces mutual flux. Multi-coil arrays and ferrite shielding mitigate this.
- EMI: Alternating fields induce stray currents. Ferrite/aluminum shielding is standard.
- Heat: Skin/proximity effects cause losses; core materials must be managed.

Example: A smartphone charger with $L_1 = L_2 = 10 \mu\text{H}$, $M = 0.5 \mu\text{H}$ ($k = 0.05$), $f = 200 \text{ kHz}$ gives $V_2 \approx 0.6 \text{ V}$ without resonance. With resonance ($Q \approx 100$), voltage boosts to $\sim 60 \text{ V}$, then regulated down. This is why all modern WPT systems use resonant compensation.

5. Conclusion

Inductive WPT applies magnetostatic and electrodynamic principles: the magnetic field carries energy via mutual inductance and Faraday's Law. Despite distance and misalignment limits, resonant techniques enable practical systems from millimeters to hundreds of millimeters and milliwatts to kilowatts. Understanding M , k , Q , and frequency is essential for efficient, safe design.

References

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